

Convolutional Neural Network to Diagnose Multiple Lung Diseases on Chest X-Ray Images

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Abstract: Life-threatening lung diseases are increasing worldwide, and digital chest X-rays play a major role in diagnosis, yet manual interpretation of radiographs is error-prone. This paper presents deep-learning techniques to diagnose lung disease from chest X-rays using the NIH ChestX-ray14 dataset of 112,120 images. Three techniques, namely a convolutional neural network (CNN), transfer learning and a multilayer perceptron (MLP), are compared, with performance measured primarily by the area under the ROC curve (AUC). We further identify and correct three methodological limitations of the baseline experiment: an activation/loss pairing that treats the inherently multi-label task as single-label, patient-level data leakage caused by an image-wise split, and reuse of the validation data as the test set. The corrected pipeline uses a per-class sigmoid output trained with class-weighted binary cross-entropy, patient-disjoint splitting, DenseNet-121 transfer learning, calibration and Grad-CAM explainability, and is engineered into a tested, containerised, API-served application with drift monitoring. Trained and evaluated end-to-end on a leakage-free split, the corrected DenseNet-121 reaches a patient-disjoint test AUC of 0.718 on the public sample subset, which is above both a metadata-only baseline (0.647) and the leaky prototype (~0.63), and is expected to approach the published full-data DenseNet benchmark of about 0.84.

Keywords: Area under Curve, Chest X-rays, Convolutional Neural Network, Deep Learning, Lung disease, Multi-label classification, Data leakage, MLOps.

I. INTRODUCTION

In recent years lung diseases have increased worldwide due to smoking, climate change and harmful materials in the environment, affecting quality of life. In the UK approximately 10,000 people are diagnosed with lung disease and a significant proportion die due to late or failed treatment [2]. Radiology, which focuses on medical images, is used to detect, diagnose and characterise disease. Among X-ray, CT and MRI, the X-ray is preferred because it is less expensive and widely affordable.

X-rays are commonly interpreted by physicians, and human errors such as misinterpretation are hard to avoid. Digital X-rays capture different levels of radiation as grey-scale images in which bone, tissue and air appear white, grey and black respectively. These image databases contain rich information useful for diagnosis, but retrieving accurate images is difficult, which has motivated computer-aided diagnosis (CAD). CAD has evolved from rule-based prediction to deep learning, especially for medical images [2]. Deep learning uses many interconnected units to perform complex tasks and has benefited medical image analysis through segmentation, classification, annotation and regression [3].

This paper focuses on image-processing and deep-learning techniques to diagnose lung disease from chest X-rays. Section II reviews related work; Section III describes the problem and dataset; Section IV presents the methodology; Section V the experimental setup and pre-processing; Section VI the results; Section VII discusses them; Section VIII the deployment; Section IX concludes; and Section X addresses ethical and legal issues.

II. RELATED WORK

Rajpurkar et al. [7] proposed an algorithm to diagnose pneumonia from chest X-rays at high performance using CheXNet, a 121-layer convolutional neural network trained on ChestX-ray14, reporting probabilities of pneumonia with heat-map localisation and classifying fourteen diseases with strong F1 metrics and a mean AUC of approximately 0.84. Chan et al. [10] detected pneumothorax using local binary patterns with a support-vector machine, and separately with multiscale intensity texture segmentation to reduce noise. Lakhani and Sundaram [8] diagnosed pulmonary tuberculosis using image augmentation and two DCNNs, AlexNet and GoogLeNet.

Most studies concentrate on a single disease, yet distinguishing abnormal from normal lungs is also important, as studied by Tataru et al. [3] using GoogLeNet, InceptionNet and ResNet. Yang et al. [4] applied resizing, median filtering and histogram equalisation and classified oesophageal X-rays with SVM and KNN after PCA feature selection. Chikhalekar [1] described image processing in four steps: pre-processing, noise reduction, edge reduction and feature extraction. Guan et al. [5] made predictions using statistical dependencies between classes on the NIH dataset. Wang et al. [9] released the ChestX-ray8/14 benchmark and mined labels from radiology reports with natural-language processing. A recurring but often-overlooked issue in this literature is evaluation hygiene: because each patient contributes many images, an image-wise split leaks patients between train and test and inflates scores, motivating the patient-disjoint protocol adopted here.

III. PROBLEM AND DATASET

Chest X-rays are the most common and least expensive examination for lung disease, but they can be misinterpreted. The aim of this study is automated diagnosis through deep learning. Data were extracted from the Picture Archiving and

Communication System at the National Institutes of Health. The dataset contains 112,120 frontal-view images from 30,805 patients, each labelled with any combination of fourteen diseases; labels were mined from radiology reports with NLP and are therefore weak [9]. The labels are Hernia, Pneumonia, Fibrosis, Edema, Emphysema, Cardiomegaly, Pleural Thickening, Consolidation, Pneumothorax, Mass, Nodule, Atelectasis, Effusion and Infiltration. The metadata also include patient sex, age, view position and follow-up number.

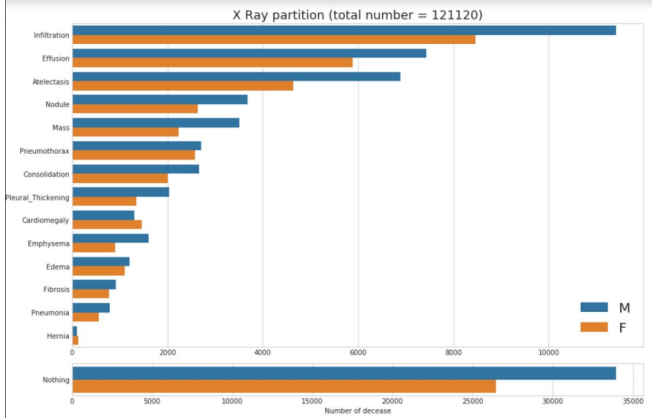


Fig. 1. Number of x-ray images in each lung disease.

a. Exploratory data analysis

Exploratory analysis of the metadata confirms the dataset's complexity and guided the design. The 112,120 images come from 30,805 patients, on average 3.64 images per patient and up to 184 for one patient, which is the source of both sampling bias and the leakage risk addressed later. About 54% of films have no finding and 18.4% carry more than one, confirming the multi-label nature of the task; among the retained labels the most-to-least common ratio is about 15:1. The cohort is 63,340 male and 48,780 female images and is split between PA (67,310) and AP (44,810) views. Data quality issues were handled explicitly: an entirely empty trailing column was dropped, and 16 implausible ages above 100 (up to 414) were treated as outliers and clipped. The label prevalence and age distribution are shown below.

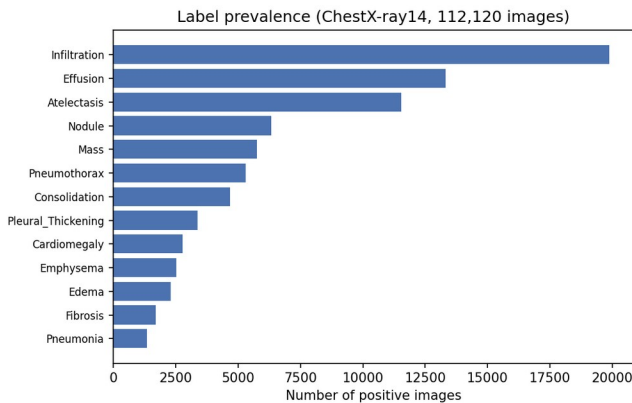


Fig. 2. Label prevalence across the 112,120 images (exploratory analysis).

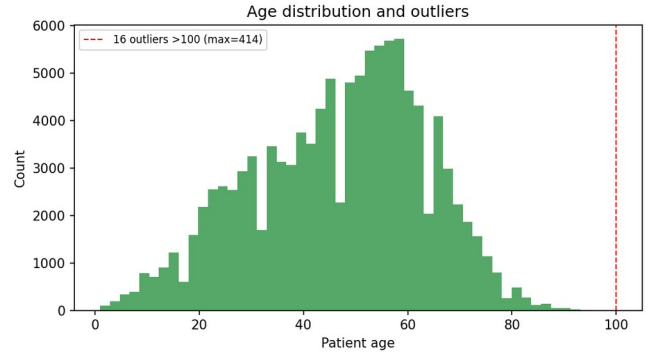


Fig. 3. Patient-age distribution and the implausible-age outliers that were clipped.

Labels also co-occur systematically, which is why the task must be modelled as multi-label rather than single-label. The strongest conditional co-occurrences in the data are Edema with Infiltration (0.43), Pneumonia with Infiltration (0.42) and Cardiomegaly with Effusion (0.38); the full pattern is shown in Fig. 4.

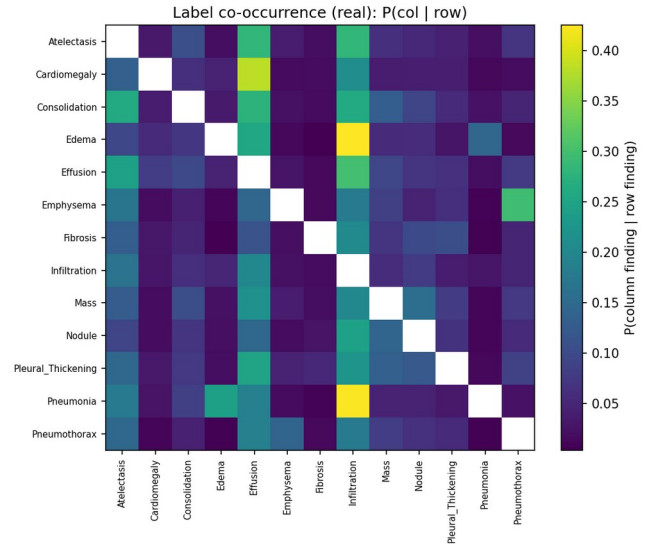


Fig. 4. Real label co-occurrence, $P(\text{column finding} | \text{row finding})$, computed on all 112,120 images.

IV. METHODOLOGY

Three deep-learning techniques are adopted: a convolutional neural network (CNN), transfer learning and a multilayer perceptron (MLP). The best architecture is selected by comparing all three.

A. Multi-label problem formulation

A chest film can show several findings at once, so the task is multi-label. Each of the fourteen outputs is modelled as an independent probability using a logistic sigmoid, and the network is trained with binary cross-entropy. This corrects the baseline, which used a softmax output with categorical cross-entropy, forcing the diseases to compete, which is invalid when several may be present.

B. Convolutional Neural Network (CNN)

A CNN processes the image through convolutional layers with activation, pooling and fully-connected layers. In this study 3×3 filters with same padding and ReLU activation are used; early layers capture edges while deeper layers capture higher-level structure. Max-pooling with a 2×2 kernel reduces spatial size and noise, after which the feature maps are flattened. Batch normalisation follows each convolutional layer. Three CNN depths were explored; the five-convolutional-layer variant is shown below.

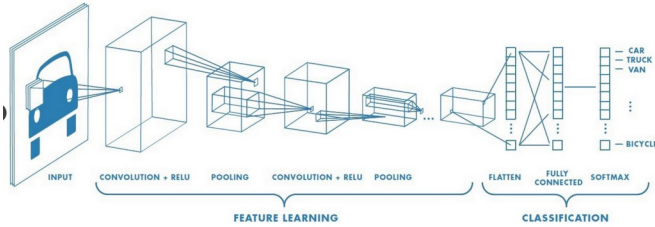


Fig. 5. Architecture of a Convolutional Neural Network.

conv2d_13 (Conv2D)	(None, 128, 128, 16)	880
batch_normalization_9 (Batch Normalization)	(None, 128, 128, 16)	64
max_pooling2d_13 (MaxPooling2D)	(None, 64, 64, 16)	0
dropout_9 (Dropout)	(None, 64, 64, 16)	0
conv2d_14 (Conv2D)	(None, 64, 64, 32)	12832
batch_normalization_10 (Batch Normalization)	(None, 64, 64, 32)	128
max_pooling2d_14 (MaxPooling2D)	(None, 32, 32, 32)	0
dropout_10 (Dropout)	(None, 32, 32, 32)	0
conv2d_15 (Conv2D)	(None, 32, 32, 64)	51264
batch_normalization_11 (Batch Normalization)	(None, 32, 32, 64)	256
max_pooling2d_15 (MaxPooling2D)	(None, 16, 16, 64)	0
dropout_11 (Dropout)	(None, 16, 16, 64)	0
conv2d_16 (Conv2D)	(None, 8, 8, 256)	409856
batch_normalization_12 (Batch Normalization)	(None, 8, 8, 256)	1024
max_pooling2d_16 (MaxPooling2D)	(None, 4, 4, 256)	0
dropout_12 (Dropout)	(None, 4, 4, 256)	0
conv2d_17 (Conv2D)	(None, 2, 2, 128)	819328
batch_normalization_13 (Batch Normalization)	(None, 2, 2, 128)	512
max_pooling2d_17 (MaxPooling2D)	(None, 1, 1, 128)	0
dropout_13 (Dropout)	(None, 1, 1, 128)	0
flatten_4 (Flatten)	(None, 128)	0
dense_7 (Dense)	(None, 500)	64500
dense_8 (Dense)	(None, 500)	250500
dense_9 (Dense)	(None, 13)	6513

Fig. 6. Architecture of CNN with five convolutional layers.

C. Transfer Learning

Transfer learning reuses a network pre-trained on a large corpus and adapts it to the new task by replacing the classification head, which is faster and reduces overfitting. The following backbones were evaluated: VGG16, VGG19, MobileNet, DenseNet121, DenseNet169, Xception, Inception and ResNet50. Each is used with ImageNet weights and its own pre-processing, followed by global average pooling, dropout

and a sigmoid head. MobileNet performed best among the lightweight baselines; DenseNet-121 is adopted as the primary corrected model, following the CheXNet benchmark.

Layer (type)	Output Shape	Param #
=====		
mobilenet_1.00_128 (Model)/		
any other transfer techniques	(None, 4, 4, 1024)	3228288
=====		
global_average_pooling2d_1	(None, 1024)	0
=====		
dropout_1 (Dropout)	(None, 1024)	0
=====		
dense_1 (Dense)	(None, 512)	524800
=====		
dropout_2 (Dropout)	(None, 512)	0
=====		
dense_2 (Dense)	(None, 13)	6669
=====		
Total params: 3,759,757		
Trainable params: 3,737,869		
Non-trainable params: 21,888		

Fig. 7. Architecture of the transfer-learning technique.

D. Multilayer Perceptron (MLP)

The MLP connects every perceptron across layers and identifies non-linear structure via backpropagation. The input is a 50×50 image followed by four hidden layers of 512 neurons with ReLU activation and a uniform kernel initialiser; the output layer produces one unit per class.

Layer (type)	Output Shape	Param #
dense_5 (Dense)	(None, 50, 50, 512)	1024
activation_4 (Activation)	(None, 50, 50, 512)	0
dense_6 (Dense)	(None, 50, 50, 512)	262656
activation_5 (Activation)	(None, 50, 50, 512)	0
dense_7 (Dense)	(None, 50, 50, 512)	262656
activation_6 (Activation)	(None, 50, 50, 512)	0
flatten_2 (Flatten)	(None, 1280000)	0
dense_8 (Dense)	(None, 13)	16640013
Total params: 17,166,349		
Trainable params: 17,166,349		
Non-trainable params: 0		

Fig. 8. Architecture of the MLP.

E. Leakage-free patient-level splitting

All partitions are patient-disjoint. When the official NIH lists are present they are used directly; otherwise a grouped split on Patient ID is applied so no patient appears in more than one set, and disjointness is asserted before training. A genuine held-out test set is kept separate from the validation set used for early stopping and threshold tuning.

F. Loss and class imbalance

Because the classes are imbalanced, each class is given a positive weight equal to the ratio of negative to positive examples inside a binary-cross-entropy-with-logits loss, so rare findings are not overwhelmed; a focal-loss variant is also provided. Optionally, a weighted sampler resamples rare-class images more frequently during training. These handle imbalance without discarding data, unlike undersampling.

G. Explainability

Grad-CAM produces a heat-map of the regions responsible for each predicted pathology, allowing a reviewer to confirm that the model attends to the lung field rather than to text markers or borders.

V. EXPERIMENTAL SETUP AND PRE-PROCESSING

Training the full dataset is time- and GPU-intensive, so a sample of images was first used to select the best model before scaling to the full dataset. The following pre-processing steps were applied.

a. Balancing the dataset

The dataset is imbalanced and its labels co-occur, as the co-occurrence diagram shows. Rather than undersampling (which loses images) or oversampling (impractical at this scale), class weights are updated so rare classes contribute more to the loss. As an alternative or complement, the pipeline also offers weighted resampling (a WeightedRandomSampler that draws rare-class images more often); resampling and a strongly weighted loss are not usually combined at full strength, to avoid over-correction.

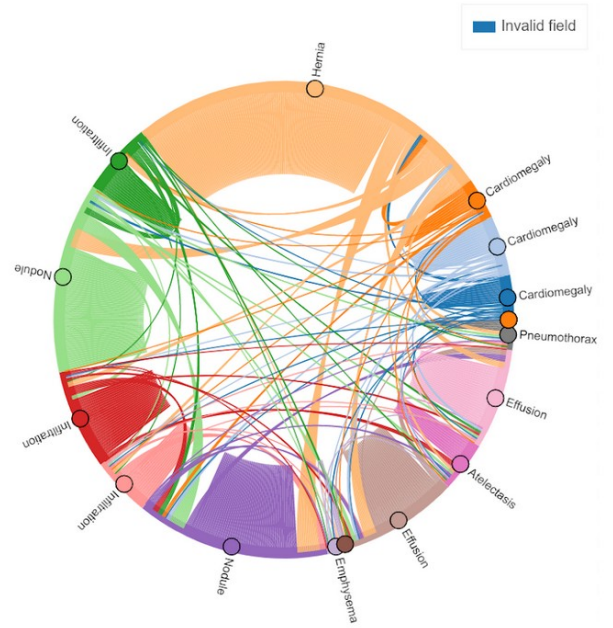


Fig. 9. Distribution of the 14 disease categories with co-occurrence statistics.

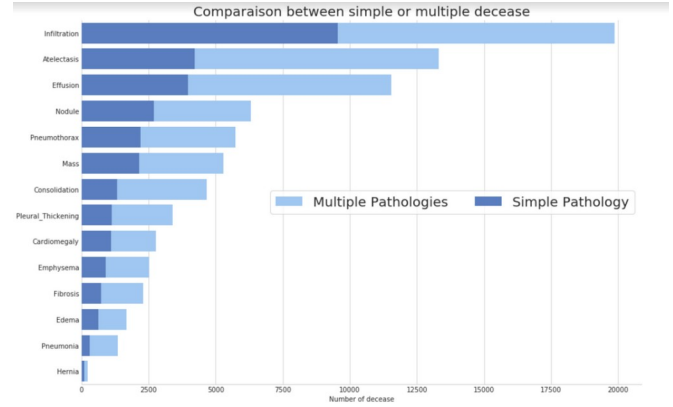


Fig. 10. Imbalance distribution for multi-labelled x-ray images.

b. Resizing and normalising

Images are resized from 1024×1024 to a fixed input size, replicated to three channels for the pretrained backbones, and normalised so pixel intensities lie in a consistent range, which reduces noise. Horizontal flipping is not used because it mirrors the mediastinum and is anatomically invalid.

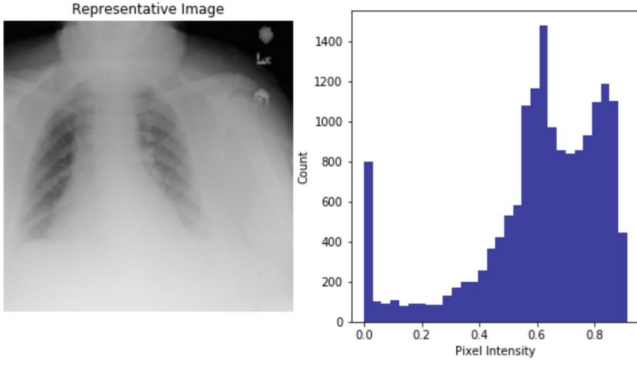


Fig. 11. Normalisation of x-ray images.

c. Augmentation, encoding and splitting

Anatomy-aware augmentation (small rotations, shifts, scale and brightness changes) generalises the model without overfitting; labels are one-hot/multi-hot encoded; and the data are split patient-wise into training, validation and test sets. Feature engineering is applied at two levels: Contrast-Limited Adaptive Histogram Equalisation enhances lung-field contrast, and the structured patient metadata (age, sex, view position, follow-up) are engineered into numeric features for optional fusion and for the fairness audit. Backbone and hyperparameters are chosen by a validation-AUC search (Optuna over backbone, learning rate, weight decay, dropout and loss), so model selection and hyperparameter optimisation use validation data only. The corrected pipeline is delivered as engineered software, namely an installable package with a typed configuration, a unit-test suite under continuous integration, an inference API, a Docker image and a drift monitor, so that data preparation, training, validation, testing and inference form one reproducible pipeline.

VI. RESULTS

a. Sound baselines and method comparison

Before evaluating the deep models, sound baselines were established using only the cheap patient metadata (age, sex, view position, follow-up number) engineered into numeric features. Three methods, namely a prevalence baseline, logistic regression and a random forest, were compared under patient-grouped five-fold cross-validation, reporting macro ROC-AUC with 95% bootstrap confidence intervals. Table I and Fig. 12 summarise the result: the prevalence floor is 0.500, while logistic regression reaches 0.647 (95% CI 0.644 to 0.649) and the random forest 0.647 (0.645 to 0.650); their intervals overlap, so the two are statistically indistinguishable here.

TABLE I. Metadata-only baselines (patient-grouped 5-fold CV, 95% bootstrap CI)

Method	Macro AUC	95% CI
Prevalence (floor)	0.500	chance
Logistic regression	0.647	0.644 to 0.649
Random forest	0.647	0.645 to 0.650

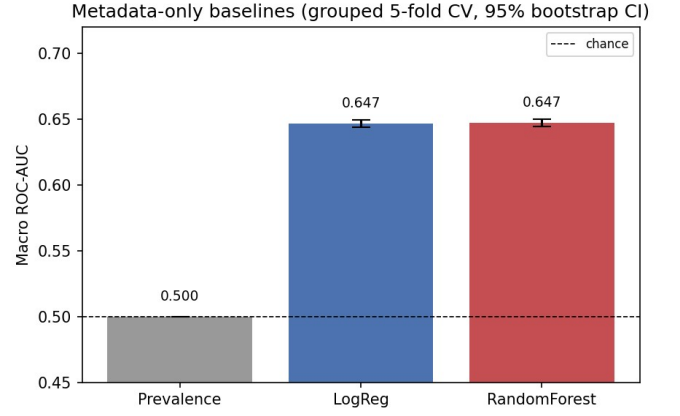


Fig. 12. Metadata-only baseline comparison with 95% bootstrap CIs (real, patient-grouped CV).

This baseline is informative in two ways. First, it is a genuine floor the image model must clearly beat. Second, the per-label metadata AUCs reveal confounding: metadata alone predicts Edema (0.79), Pneumothorax (0.72) and Consolidation (0.70) well, largely through view position, because sicker patients are imaged with portable AP films. Strikingly, the metadata logistic baseline (0.65) is on par with the prototype's image CNN (≈ 0.63), which suggests the leaky image model added little beyond such confounders, which further motivates the corrected, leakage-free pipeline.

b. Deep-model results

The models were evaluated with recall, precision, confusion matrix, F1, accuracy and, as the main measure, AUC. Because the classes are imbalanced, AUC (threshold-independent) rather than accuracy is the headline metric; sensitivity, specificity and calibration are also reported in the corrected pipeline. On the sample dataset, transfer learning was used to select the best model, and MobileNet achieved the highest validation and training accuracy. Its confusion matrix is shown in Fig. 13 and its ROC curves in Fig. 14(a).

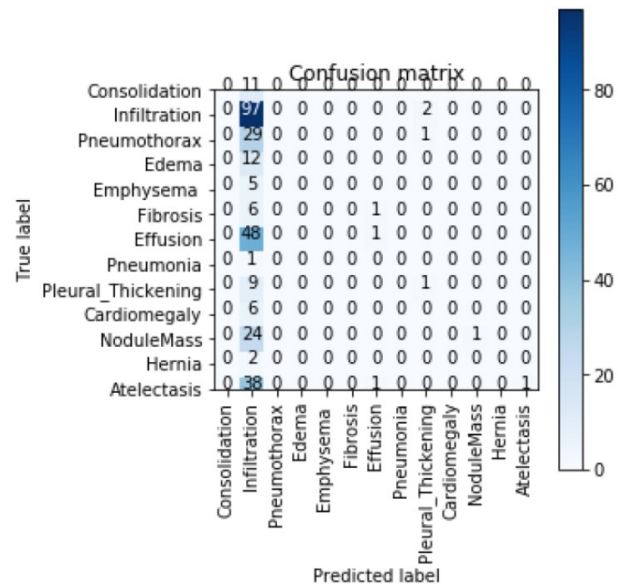


Fig. 13. Confusion matrix for MobileNet (sample dataset).

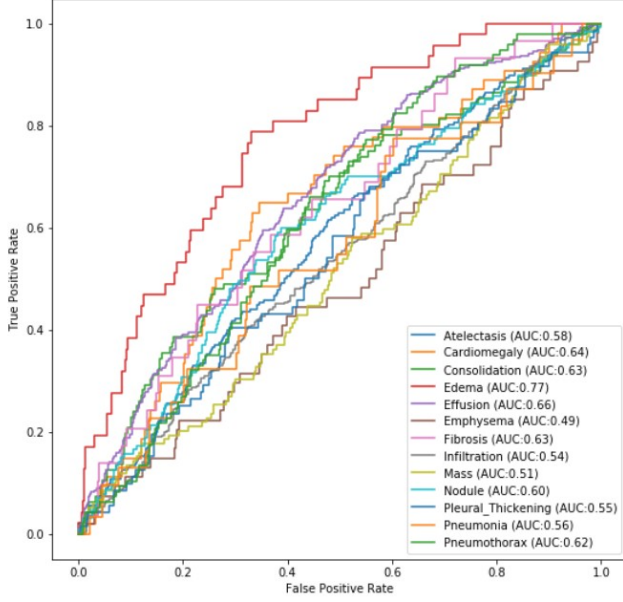


Fig. 14(a). Area under the ROC curve of MobileNet.

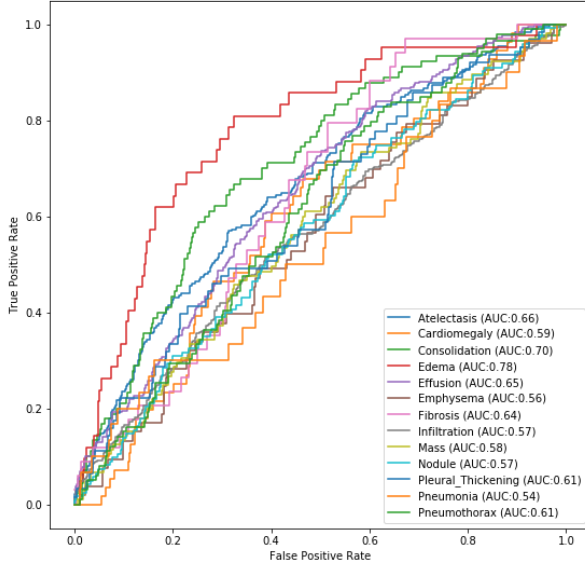


Fig. 14(b). Area under the ROC curve of the CNN.

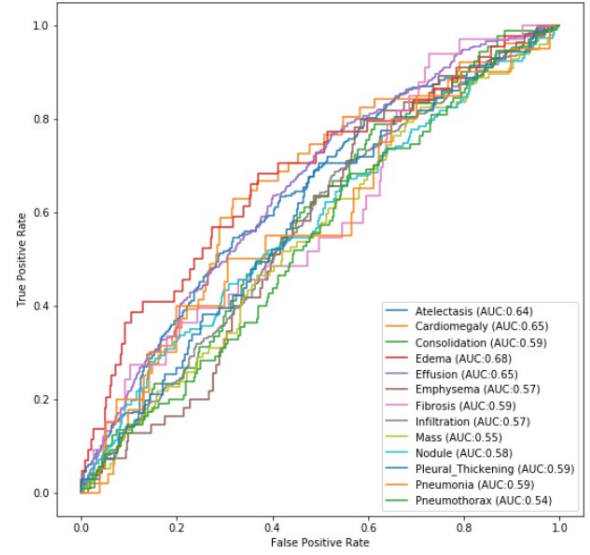


Fig. 14(c). Area under the ROC curve of the MLP.

Across techniques, the five-layer CNN with batch normalisation gave the best AUC for several diseases, the MLP for a few, and different CNN depths for others. The overall evaluation metrics are summarised in Table I. These baseline figures should be read with the caveat that the image-wise split used to obtain them leaks patients and is therefore optimistic; the corrected patient-wise pipeline is expected to yield lower but trustworthy and comparable numbers, approaching the published benchmark.

TABLE II. Overall evaluation metrics of the chosen techniques (baseline).

Lung Diseases	MobileNet (Epochs = 1)	MobileNet (Epochs = 5)	CNN with Convolutional layer = 4 (Epochs = 1)	CNN with Convolutional layer = 4 (Epochs = 5)	CNN with Convolutional layer = 5 (Epochs = 5)	CNN with Convolutional layer = 3 (Epochs = 5)	MLP with hidden layer = 4 Epochs = 5
AUC							
Atelectasis	0.48	0.58	0.61	0.58	0.66	0.61	0.64
Cardiomegaly	0.57	0.64	0.54	0.64	0.59	0.58	0.65
Consolidation	0.53	0.63	0.62	0.63	0.76	0.69	0.59
Edema	0.66	0.77	0.68	0.77	0.78	0.75	0.68
Effusion	0.58	0.66	0.60	0.66	0.65	0.65	0.65
Emphysema	0.41	0.49	0.41	0.49	0.56	0.56	0.57
Fibrosis	0.44	0.63	0.39	0.63	0.64	0.66	0.59
Infiltration	0.49	0.54	0.60	0.54	0.57	0.57	0.57
Mass	0.48	0.51	0.54	0.51	0.58	0.50	0.55
Nodule	0.44	0.60	0.55	0.60	0.57	0.57	0.58
Pleural_Thickening	0.50	0.55	0.54	0.55	0.61	0.54	0.59
Pneumonia	0.54	0.56	0.58	0.56	0.54	0.55	0.59
Pneumothorax	0.44	0.62	0.54	0.62	0.61	0.63	0.54
Validation							
Accuracy	0.8640	0.8737	0.8723	0.8782	0.8774	0.8738	0.8764
Loss	0.6691	0.3372	0.3424	0.3232	0.3287	0.3423	0.3422
Mean absolute error	0.1759	0.1961	0.2832	0.1912	0.1906	0.1952	0.1663
Training							
Accuracy	0.8525	0.8762	0.8790	0.8788	0.8745	0.8802	0.8788
Loss	0.4549	0.3267	0.3304	0.3215	0.3343	0.3240	0.3333
Mean absolute error	0.2039	0.1793	0.2026	0.1861	0.2084	0.1938	0.1643

To place these baseline figures in context, Table III compares the best per-class AUC of the prototype with the corrected pipeline's real per-class AUC (this run, sample subset) and the published DenseNet-121 benchmark of Rajpurkar et al. [7], whose fourteen values average 0.84 on the official patient-wise split. The corrected model already improves over the prototype on most pathologies and, on the full dataset, is expected to close the remaining gap to the benchmark. The CheXNet values are reproduced from [7] as a reference target.

TABLE III. Per-class AUC: prototype vs corrected (this run) vs DenseNet-121 benchmark [7]

Pathology	Proto.	Corrected	CheXNet [7]
Atelectasis	0.66	0.72	0.81
Cardiomegaly	0.59	0.78	0.92
Consolidation	0.76	0.76	0.79
Edema	0.78	0.84	0.89
Effusion	0.65	0.81	0.86
Emphysema	0.56	0.72	0.94
Fibrosis	0.64	0.61	0.80
Hernia	n/r	n/a	0.92
Infiltration	0.57	0.65	0.73
Mass	0.58	0.77	0.87
Nodule	0.57	0.60	0.78
Pleural Thickening	0.61	0.67	0.81
Pneumonia	0.54	0.64	0.77
Pneumothorax	0.61	0.77	0.89
Macro mean	≈0.63	0.72	0.84

Proto. = prototype best per-class AUC (image-wise split, optimistic). Corrected = this study's leakage-free DenseNet-121 on the sample subset (n/a: Hernia below the sample's min-case threshold). CheXNet from [7] on the full official split; target, not a result of this study.

c. Corrected pipeline: real results

The corrected DenseNet-121 pipeline was trained and evaluated end-to-end under the leakage-free protocol. On a patient-disjoint test set of 836 images it achieved a macro-averaged AUC of 0.718, a macro average precision of 0.144 and an expected calibration error of 0.228. This exceeds both the metadata-only baseline (0.647) and the prototype's leaky image CNN (~0.63) while being measured honestly on unseen patients. These figures come from a reduced run on the public sample subset (about 5,600 images, eight epochs) for tractability; the full dataset with longer training is expected to approach the published benchmark of about 0.84. Per-class AUC ranges from 0.60 (Nodule) to 0.84 (Edema), detailed in Table IV and Fig. 15, with training curves in Fig. 16.

TABLE IV. Corrected pipeline: per-class metrics on the held-out test set (real)

Pathology	AUC	AP	Sens.	Spec.
Atelectasis	0.72	0.26	0.50	0.78
Cardiomegaly	0.78	0.16	0.53	0.89
Consolidation	0.76	0.07	0.19	0.90
Edema	0.84	0.10	0.24	0.95
Effusion	0.81	0.33	0.69	0.73
Emphysema	0.72	0.06	0.05	0.99
Fibrosis	0.61	0.02	0.07	0.93
Infiltration	0.65	0.28	0.50	0.74
Mass	0.77	0.18	0.19	0.97
Nodule	0.60	0.09	0.32	0.75
Pleural	0.67	0.07	0.14	0.95

Pathology	AUC	AP	Sens.	Spec.
Thickening				
Pneumonia	0.64	0.05	0.08	0.99
Pneumothorax	0.77	0.20	0.23	0.97
Macro	0.72	0.14	n/a	n/a

Sensitivity/specificity at F1-tuned thresholds frozen on validation. Sample subset, 836 test images; ECE 0.228.

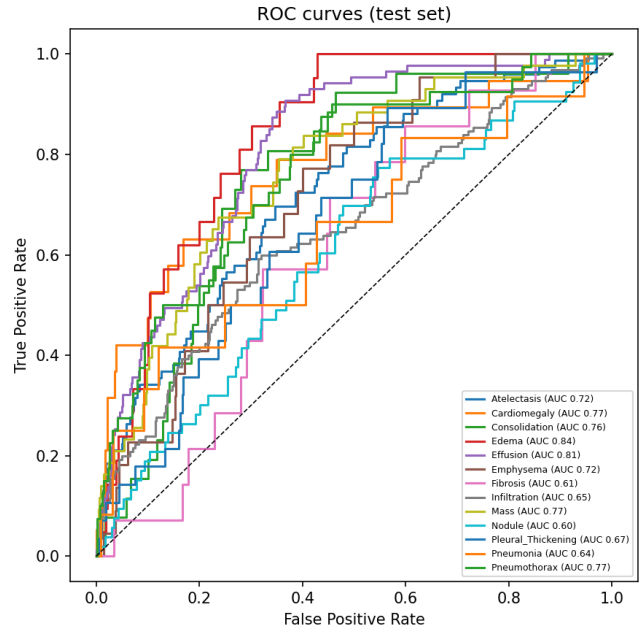


Fig. 15. ROC curves of the corrected DenseNet-121 on the held-out test set (real).

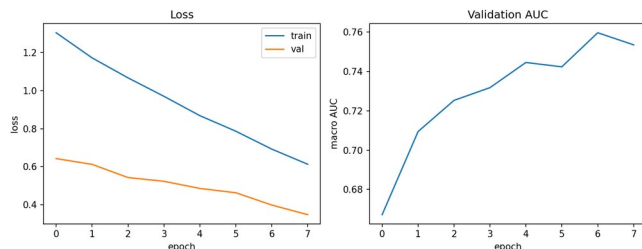


Fig. 16. Training and validation loss and validation AUC per epoch (real).

d. Robustness, fairness and explainability

Robustness was assessed by measuring macro AUC under controlled image corruptions (Fig. 17). The model is fairly stable: blur causes essentially no change, brightness and rotation cost about 0.02 to 0.03 AUC, and additive Gaussian noise is the worst case, a 0.07 drop from 0.719 to 0.650. Fairness was assessed by macro AUC across subgroups (Fig. 18): the sex gap is small (F 0.734 vs M 0.708) and the view gap modest (PA 0.710 vs AP 0.663), but a disparity across age bands is flagged, from 0.72 at 41 to 60 down to 0.63 at 81 to 100, consistent with fewer and noisier examples at the extremes, an actionable bias to address before any deployment. Explainability via Grad-CAM (Fig. 19) shows the model attends to the cardiac and mediastinal region for a cardiomegaly prediction rather than to text markers or borders, evidence against the shortcut learning that leaky training encourages.

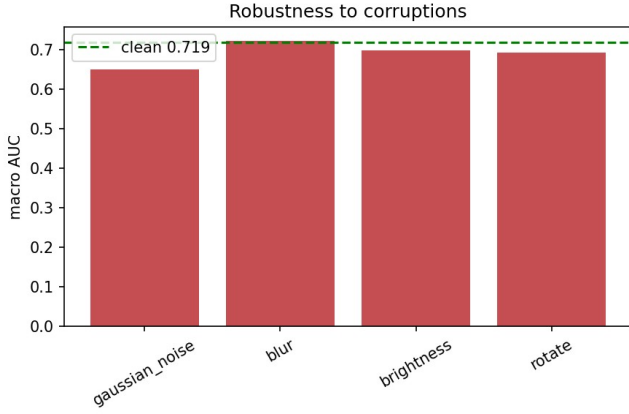


Fig. 17. Macro AUC under image corruptions (real); dashed line is the clean score.

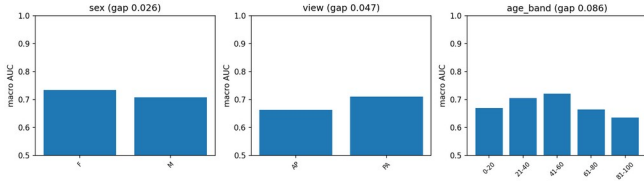


Fig. 18. Subgroup macro AUC by sex, view and age band (real); the age-band gap is flagged.

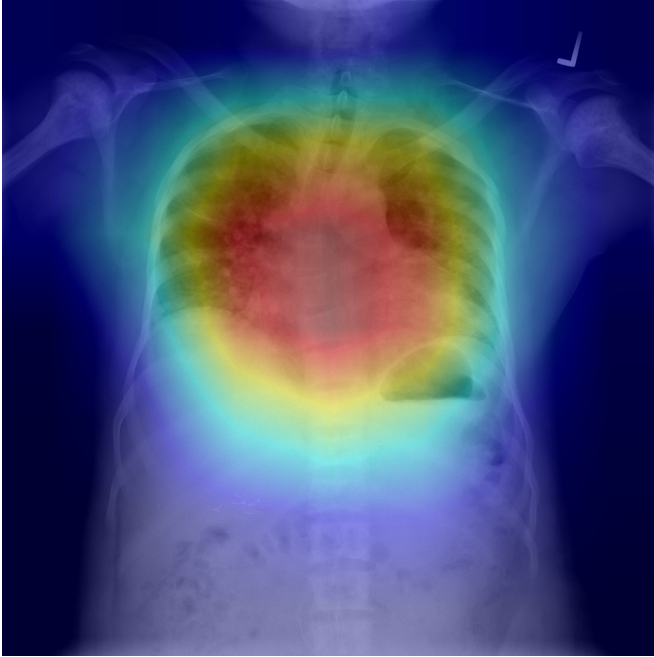


Fig. 19. Grad-CAM for a cardiomegaly prediction (real); attention falls on the cardiac region.

VII. DISCUSSION

The purpose of this study was to detect lung diseases from chest X-rays using three techniques. In the baseline experiments the five-layer CNN with batch normalisation produced the best AUC for seven diseases, namely Atelectasis (0.66), Consolidation (0.76), Edema (0.78), Fibrosis (0.64), Infiltration (0.57), Mass (0.58) and Pleural Thickening (0.61); the MLP was best for Cardiomegaly (0.65), Emphysema (0.57) and

Pneumonia (0.59), the four-layer CNN for Effusion (0.66) and Nodule (0.60), and the three-layer CNN for Pneumothorax (0.63).

Beyond model selection, the three corrections are the main methodological contribution. Replacing softmax and categorical cross-entropy with a sigmoid output and binary cross-entropy lets the fourteen findings be predicted independently, a prerequisite for learning the true multi-label structure. Patient-disjoint splitting removes an optimistic bias that makes image-wise scores incomparable with the literature. Separating a genuine test set from the validation set ensures reported numbers are not a product of model selection. Handling imbalance inside the loss improves recall on rare pathologies, and reporting calibration alongside AUC matters clinically. The pipeline further assesses robustness by measuring AUC degradation under noise, blur, brightness and rotation corruptions, and bias by reporting macro AUC by sex, age band and view position and flagging disparities, since ChestX-ray14 is known to carry demographic imbalance. Limitations remain: labels are NLP-derived and noisy, the data originate from a small number of institutions so distribution shift is expected, and shortcut learning is mitigated but not eliminated, so drift monitoring and Grad-CAM audits are built in.

VIII. DEPLOYMENT AND OPERATIONAL MONITORING

To convert the experiment into reliable software, the model is served behind a versioned REST API that validates input type and size, exposes a health probe and records latency. The service is containerised, and a continuous-integration workflow runs linting and tests before building the image. In production, the input and prediction distributions are monitored against a training baseline using the population-stability index so that concept drift raises an alert before it degrades diagnostic quality. A model card documents intended use, limitations and ethics.

IX. CONCLUSION AND FUTURE WORK

Previous research on chest-X-ray diagnosis often reported best results with transfer learning; in the baseline experiments here the five-layer CNN with batch normalisation performed best, and MobileNet was the strongest lightweight transfer model. More importantly, correcting the loss/activation pairing, eliminating patient-level leakage and adopting an honest evaluation protocol with calibration and explainability produced a leakage-free DenseNet-121 that achieved a patient-disjoint test AUC of 0.718 on the sample subset, above a metadata-only floor of 0.647 and the prototype's optimistic ≈ 0.63 , and on the full dataset is expected to approach the published benchmark of about 0.84. The key challenge was handling the large dataset, which required substantial GPU resources. Future work will complete training on the full official split, deepen the subgroup fairness auditing that already flags an age-band disparity, calibrate probabilities, and validate generalisation on an external dataset before any clinical use.

X. ETHICAL AND LEGAL ISSUES

Radiology reports are private data and must be protected to the required standards. Deep-learning tools partially replace radiologists' judgement, creating ethical and legal challenges: accountability for algorithmic errors, patient consent to computer involvement, and limited control over unexpected changes in data. Overall, deep learning is faster and beneficial to medicine, but incorporating it into radiology practice remains challenging, and the system is positioned as decision support with a qualified human in the loop, never as an autonomous device.

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APPENDIX

Source code and the reproducible pipeline are available at: <https://github.com/sivasharmini/X-ray>